

A Counterexample and a Positive Non-Extreme Theorem for Composed de Branges–Rovnyak Spaces

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Abstract

Dellepiane and Seco ask whether every analytic self-map ψ of the disk induces a bounded inclusion $C_\psi \mathcal{H}(b) \subseteq \mathcal{H}(b \circ \psi)$ for every Schur function b . We give a two-dimensional-data counterexample: $b(z) = z$ and any disk automorphism not fixing zero. We also prove that C_ψ is contractive for arbitrary b when $\psi(0) = 0$, and is bounded for arbitrary ψ whenever b is non-extreme. Thus the problem as stated has a negative answer, while its motivating non-extreme case has a positive answer.

1 Source problem

The source is Dellepiane and Seco’s *Orthogonal polynomials in de Branges–Rovnyak spaces* [1]. It is arXiv:2603.03093v1. Problem 3.11 on PDF page 13 asks whether one always has

$$C_\psi \mathcal{H}(b) \subseteq \mathcal{H}(b \circ \psi)$$

and, if so, whether $C_\psi : \mathcal{H}(b) \rightarrow \mathcal{H}(b \circ \psi)$ is bounded. Here is the complete source passage at full readable page width.

3.4. Further remarks. Theorem 3.9 and its proof raise some interesting questions:

Problem 3.10. *Given a more general function $\psi : \mathbb{D} \rightarrow \mathbb{D}$, and $b \in H_1^\infty$, what can be said about the de Branges–Rovnyak space $H(C_\psi b)$ and its relation with $H(b)$?*

Our analysis provides rather precise answers in the case where $\psi(z) = z^N$ (and b is non-extreme), but this matter can clearly be studied in greater generality. We propose two further problems. The first one is about the bounded embedding:

Problem 3.11. *Is it always true that $C_\psi H(b) \subseteq H(C_\psi b)$ and if so, is*

$$C_\psi : H(b) \rightarrow H(C_\psi b)$$

a bounded operator?

The second is perhaps even more ambitious:

Problem 3.12. *Is there a way to express $H(C_\psi b)$ in terms of $H(b)$?*

2 Definitions and kernel criterion

For a Schur function b , the de Branges–Rovnyak space $\mathcal{H}(b)$ is the reproducing-kernel Hilbert space on $\mathbb{D}$ with kernel

$$K^b(z, w) = \frac{1 - b(z)\overline{b(w)}}{1 - z\overline{w}}.$$

For an analytic self-map $\psi : \mathbb{D} \rightarrow \mathbb{D}$, write $C_\psi f = f \circ \psi$.

Lemma 1 (Composition-kernel criterion). *The operator $C_\psi : \mathcal{H}(b) \rightarrow \mathcal{H}(b \circ \psi)$ has norm at most M if and only if the kernel*

$$M^2 K^{b \circ \psi}(z, w) - K^b(\psi(z), \psi(w))$$

is positive semidefinite.

Proof. On reproducing kernels the formal adjoint must satisfy

$$C_\psi^* K^{b \circ \psi}(\cdot, w) = K^b(\cdot, \psi(w)).$$

The inequality $\|C_\psi^* g\|^2 \leq M^2 \|g\|^2$ on finite linear combinations of kernel functions is exactly positivity of the displayed kernel. Density then gives the equivalence and constructs the bounded operator in the positive-kernel direction. $\square$

We will repeatedly use the elementary factorization

$$K^{b \circ \psi}(z, w) = K^b(\psi(z), \psi(w)) K^\psi(z, w). \quad (1)$$

3 Proof intuition

The factor K^ψ in (1) is the whole issue. If $\psi(0) = 0$, Schwarz’s lemma writes $\psi(z) = zq(z)$ with q Schur, and then $K^\psi - 1 = z\overline{w}K^q$ is positive. The kernel criterion therefore gives contractivity with no assumption on b .

For a general ψ , a disk automorphism moves $\psi(0)$ to zero. Weighted composition by that automorphism is unitary between the appropriate de Branges–Rovnyak spaces. Removing its elementary weight requires multiplication by a linear polynomial. That multiplication is bounded in the non-extreme case because the forward shift is bounded there.

The same automorphism reveals why the unrestricted assertion fails. For $b(z) = z$, the source space consists only of constants. After composition with a non-origin automorphism, the target is a different one-dimensional space, generated by a nonconstant Cauchy kernel, and therefore does not contain the constant function one.

4 The result

Theorem 1. *Let b be a Schur function and $\psi : \mathbb{D} \rightarrow \mathbb{D}$ analytic.*

1. *If $\psi(0) = 0$, then*

$$C_\psi : \mathcal{H}(b) \longrightarrow \mathcal{H}(b \circ \psi)$$

is a contraction.

2. *If b is non-extreme in the closed unit ball of H^∞ , then the same operator is bounded for every ψ .*

3. Without non-extremality the inclusion can fail. More precisely, for any $a \in \mathbb{D} \setminus \{0\}$, put

$$b(z) = z, \quad \psi_a(z) = \frac{a - z}{1 - \bar{a}z}.$$

Then $1 \in \mathcal{H}(b)$, but

$$C_{\psi_a} 1 = 1 \notin \mathcal{H}(b \circ \psi_a).$$

Consequently Problem 3.11 has a negative answer as stated.

Proof. Suppose first that $\psi(0) = 0$. By Schwarz's lemma, $\psi(z) = zq(z)$ for a Schur function q . Hence

$$K^\psi(z, w) - 1 = z\bar{w} \frac{1 - q(z)\overline{q(w)}}{1 - z\bar{w}} = z\bar{w}K^q(z, w),$$

which is a positive kernel. Using (1),

$$K^{b \circ \psi}(z, w) - K^b(\psi(z), \psi(w)) = K^b(\psi(z), \psi(w))(K^\psi(z, w) - 1)$$

is positive by the Schur product theorem for positive-kernel Gram matrices. Lemma 1 with $M = 1$ proves part 1.

For part 2, set $\alpha = \psi(0)$ and introduce the involutive disk automorphism

$$\varphi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

Put $u = \varphi_\alpha \circ \psi$, so $u(0) = 0$ and $\psi = \varphi_\alpha \circ u$. Also put $\tilde{b} = b \circ \varphi_\alpha$. The automorphism identity gives

$$K^{\tilde{b}}(z, w) = h_\alpha(z)\overline{h_\alpha(w)}K^b(\varphi_\alpha(z), \varphi_\alpha(w)), \quad h_\alpha(z) = \frac{\sqrt{1 - |\alpha|^2}}{1 - \bar{\alpha}z}. \quad (2)$$

Thus the weighted composition operator

$$U_\alpha f = h_\alpha(f \circ \varphi_\alpha)$$

is unitary from $\mathcal{H}(b)$ onto $\mathcal{H}(\tilde{b})$; this follows directly from (2) first on kernel spans and then by density.

Non-extremality is preserved under precomposition by a disk automorphism. Indeed, the boundary change-of-variables density is bounded above and below, so integrability of $\log(1 - |b|)$ is preserved. The forward shift $M_{\tilde{z}}$ is therefore bounded on $\mathcal{H}(\tilde{b})$, using the standard non-extreme shift characterization recalled in the source paper [1, p. 1]. Since

$$C_{\varphi_\alpha} = M_{1/h_\alpha} U_\alpha, \quad M_{1/h_\alpha} = \frac{I - \bar{\alpha}M_{\tilde{z}}}{\sqrt{1 - |\alpha|^2}},$$

composition by φ_α is bounded from $\mathcal{H}(b)$ to $\mathcal{H}(\tilde{b})$. Part 1 says that $C_u : \mathcal{H}(\tilde{b}) \rightarrow \mathcal{H}(\tilde{b} \circ u)$ is contractive. Finally,

$$C_\psi = C_u C_{\varphi_\alpha}, \quad \tilde{b} \circ u = b \circ \psi,$$

which proves part 2.

For part 3, take $b(z) = z$. Its kernel is

$$K^b(z, w) = \frac{1 - z\bar{w}}{1 - z\bar{w}} = 1,$$

so $\mathcal{H}(b) = \text{span}\{1\}$. Since $b \circ \psi_a = \psi_a$, another direct automorphism calculation gives

$$K^{\psi_a}(z, w) = \frac{1 - |a|^2}{(1 - \bar{a}z)(1 - a\bar{w})} = h_a(z)\overline{h_a(w)}.$$

Consequently $\mathcal{H}(b \circ \psi_a) = \text{span}\{h_a\}$. For $a \neq 0$, the function h_a is nonconstant, so the constant function one is not in this one-dimensional target space. But $C_{\psi_a}1 = 1$, proving failure of the inclusion. $\square$

5 Verification, limitations, and novelty

The proof is entirely encoded in reproducing kernels. Its three audit points are: the direction of Lemma 1; positivity of $z\bar{w}K^q$; and the removal of the automorphism weight using the bounded forward shift. The counterexample itself only uses the two displayed rank-one kernel calculations.

The negative example uses the extreme inner symbol $b(z) = z$. It does not contradict the source paper’s positive monomial-composition theorem for non-extreme symbols. Part 2 instead confirms boundedness throughout that non-extreme setting. The packet does not characterize the remaining extreme symbols and does not fully answer the broader descriptive Problems 3.10 and 3.12.

Before promotion, the run indexes were searched for arXiv:2603.03093, the paper title, *de Branges–Rovnyak composition operator*, and the exact inclusion into the composed-symbol space. Bounded web/arXiv searches used the exact Problem 3.11 wording, title/authors, and nearby variants. They found the source paper and work on different mappings involving fixed non-inner rational spaces, but no answer to this problem. The source still had only version 1. Novelty confidence is moderate because the proof is short and may be folklore; no priority claim is made.

6 Human-review recommendation

Classify this as a *likely valid counterexample with a complementary positive theorem*. A verifier should first check the two rank-one kernel computations, then Lemma 1 and the automorphism factorization in part 2. No numerical or conditional dependency remains.

References

- [1] E. Dellepiane and D. Seco, *Orthogonal polynomials in de Branges–Rovnyak spaces*, arXiv:2603.03093v1, 2026.